Asphalt Paver Handbook

For Cat® F Series Large Asphalt Pavers



Table of Contents

Important Safety Information 3

Paving by the Numbers 6

F-Series Grade and Slope System 44

Preparing to Pave (Grade) 69

Preparing to Pave (Slope) 76

Auto Calibration (Cat G/S) 79

Averaging Skis are the Keys
to Smoothness 92

Conclusion 99

Do not operate or work on machine unless you have read and understand the instruc tions and warnings in the Operator and Maintenance Manuals. Failure to follow the instructions or heed the warnings could result in injury or death. Contact any Cat Dealer for replacement manuals. Proper care is your responsibility.

Important Safety Information

Most accidents involving product operation, maintenance and repair are caused by failure to observe basic safety rules and precautions. An accident can often be avoided by recognizing potentially hazardous situations before an accident occurs. A person must be alert to potential hazards. This person should also
have the necessary training, skills and tools
to perform these functions properly.

Improper operation, lubrication, maintenance or repair of this product can be dangerous and could result in injury or death.
Do not operate or perform any lubrication, maintenance or repair on this product, until you have read and understood the operation, lubrication, maintenance and repair information.

Safety precautions and warnings are provided
in this presentation and on the product. If these hazard warnings are not heeded, bodily injury
or death could occur to you or other persons.

The hazards are identified by the “Safety Alert Symbol” and followed by a “Signal Word” such as “WARNING” as shown below.

The meaning of this safety alert symbol is
as follows:



Attention! Become Alert!
Your Safety is Involved.

(continued on page 4)

Operations that may cause product damage are identified by NOTICE labels on the product and in this presentation.

Caterpillar cannot anticipate every possible circumstance that might involve a potential hazard. The warnings in this presentation and on the product are therefore not all inclusive. If a tool, procedure,work method or operating technique not specifically recommended by Caterpillar is used, you must satisfy yourself that it is safe for you and others. You should also ensure that the product will not be damaged or made unsafe by the operation, lubrication, maintenance or repair procedures you choose.

The information, specifications, and illustrations in this presentation are on the basis of information available at the time it was written. The specifications, illustrations and other items can change at any time. These changes can affect the service given to the product. Obtain the complete and most current information available.

An asphalt paver is a machine that is used for road construction. The asphalt paver typically uses bulk material for the placement of paved surfaces. Most commonly, hot mix asphalt is supplied by vehicle delivery to the paver’s front hopper.

The asphalt paver is a heavy duty machine that can place paving surfaces for many applications. The asphalt paver can be used for numerous applications such as roads, city streets, parking lots, highways and motorways. The asphalt paver accepts delivery of asphalt mix from a truck or transport vehicle at the front. The asphalt paver then contains the mix in a hopper. Slat conveyors transport the mix from the hopper to the rear of the machine. The mix flows onto the ground ahead of the screed. The screed shapes the mix to specified dimensions. The screed also levels the mix to specified dimensions. The slope and the grade of the asphalt surface is precisely controlled by sensors and references that are mounted on the paver. This system achieves a high degree of flatness. The paver can be configured with different types of screeds to satisfy common customer's requirements.



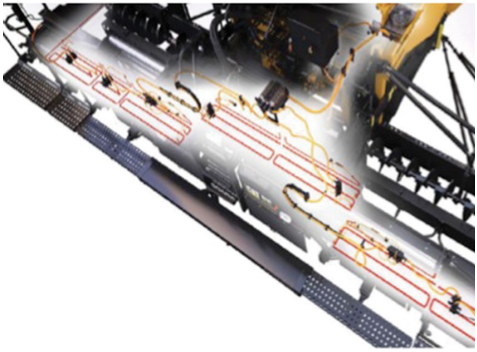
Falling off of the machine could cause serious injury or death. No riders while machine is in travel mode. Refer to Operation and Maintenance Manual, "Roading the Machine" for proper roading procedures. Always wear a seatbelt.



Producing a smooth asphalt surface with a uniform finish is not as simple as saying “one—two—three.” But a well-trained paver crew, which follows a set of established fundamental practices, will avoid many of the problems that affect mat quality.

The purpose of this presentation is to review the steps you should follow each time you get ready to pull off the joint and start paving.

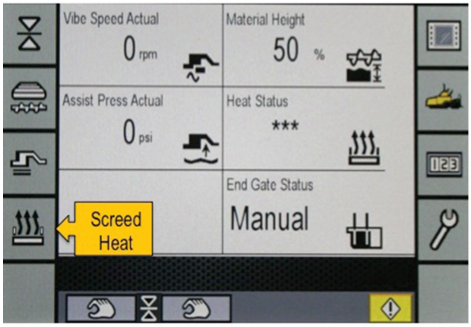
NOTE: This presentation should be used for the 3 m (10 foot) F-series paver with a SE60 V XW. This does not take the place of the Operation & Maintenance manual. Please refer to this manual for further explanation about the proper use, safety warnings and control of the machine.



STEP 1: Heat The Screed

If the screed is cold, you have to heat it before beginning to pave to prevent asphalt sticking to the screed plates. There are several types of screed heaters on Cat screeds, diesel fired, liquid propane and electric element. In this presentation, we will talk about the electric screed heat only. If you need information on the Diesel or Liquid Propane screed heat, please refer to the Operation & Maintenance Manual (O&MM) for that screed.

On Cat extendable-type screeds, each screed frame section is equipped with a heater. One on each screed extender and two on the main screed.

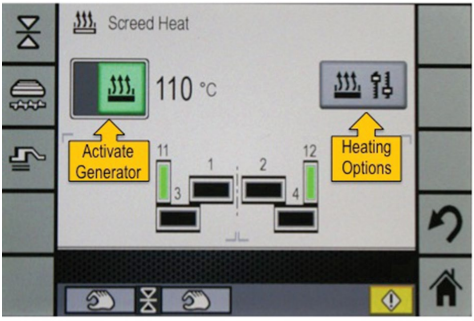


STEP 1: Heat The Screed *(Continued)*

The F-series asphalt paver has two spots to activate the heating system. This can be achieved on the left or the right tractor operator station or on the left or right screed operator station.

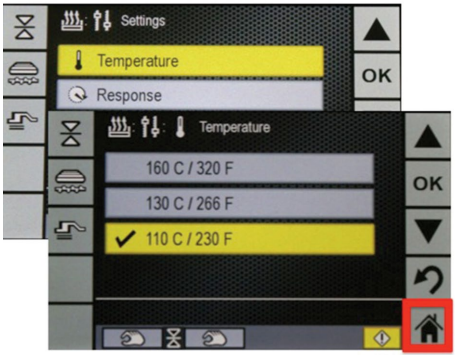
Choose the screed heat icon. (Screed operators station screen shown)

Start with the screed approximately 25-75 mm (1-3”) off of the ground. If the screed is at maximum height, wind can effect the performance of the heaters. Also if the screed is on the ground, the ground acts as a heat sink and it will take longer for the screed to come up to temperature.



STEP 1: Heat The Screed *(Continued)*

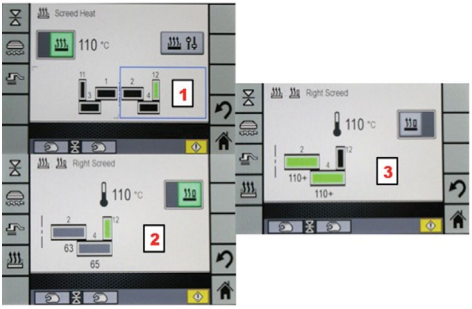
The generator icon is located on the left side of the screed heat screen. Touch the screen symbol and slide to the right. When activated the icon will turn green and the engine rpm will raise to 1300 rpm and the generator will be activated. The temperature set point can be chosen by selecting the heating option icon on the right hand side.



STEP 1: Heat The Screed *(Continued)*

In the heating options menu, the operator can choose the Temperature title. Under this screen, three temperature set points can be chosen. These set points are 110ºC (230ºF), 130ºC (266ºF), or 160ºC (320ºF). Once the temperature choice has been made, touch the Home icon and continue with the setup of "Paving by the Numbers" instruction.

Once the generator has been turned on and the temperature has been chosen the screed will heat up.



STEP 1: Heat The Screed *(Continued)*

The operator can view the status of the screed heat by touching the zone section (1). The color of the viewed zone will be black, grey (2) or green (3) (see color explanation below). Once all the zones have turned green paving can begin.

NOTE: Picture (2) shows the optional end-gate heater turned on and both the heater and the on/off button is shown in green. In picture (3), it is shown turned off and is black.

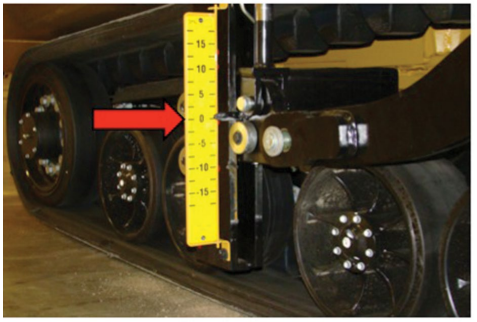
Temperatures of color changes:

Heat Setting / Black / Grey / Green

110º C (230ºF) = 0º – 59º C (32º-138ºF) / 60º - 109º C (140º-228ºF) / 110º C+ (230ºF+)

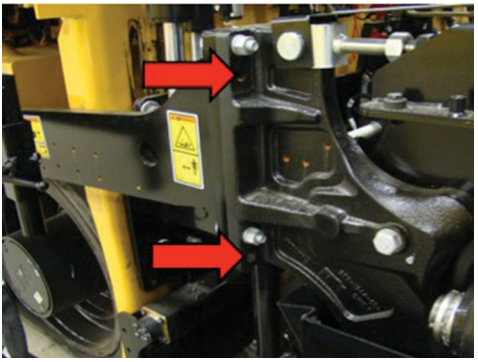
130º C (266ºF) = 0º - 79º C (32º-174ºF) / 80º - 129º C (176º-264ºF) / 130º C+ (266ºF+)

160º C (320ºF) = 0º - 109º C (32º-228ºF) / 110º - 159º C (230º-318ºF) / 130º C+ (320ºF+)



STEP 2: Position The Tow Arm Cylinder

Step 2 is positioning the tow arm cylinder for the correct ”line on pull” for the depth that is being paved. On the F-series paver, the position of the screed attachment point can be placed in a position for shallow paving 0-18 cm (0-7 in) or deep depth paving for 18 cm – 30 cm (7-12 in). For shallow paving setup, the operator should adjust the position of the tow arm pointer at “0” if paving 7.5 cm (3 in) or less. When paving at depths more than 7.5 cm (3 in), the operator should adjust the cylinder at the rate of 2.5 cm (1 in) per 2.5 cm (1 in) depth increase.

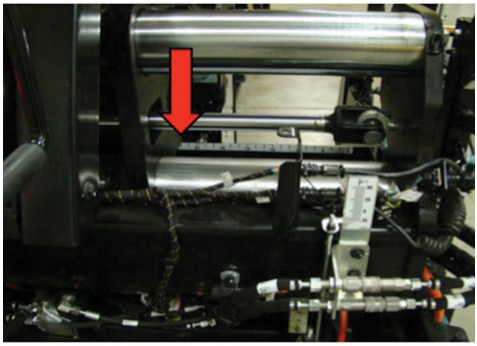


STEP 2: Position The Tow Arm Cylinder
(Continued)

Deep Depth Tow Arm Position: The F-series can be setup to pave deep depth. To do this, reposition the tow arm attachment on the drop arm to the bottom set of holes. This will give the depth range of 18 cm (7 in) – 30 cm (12 in).

When paving at this depth the tow arm will need to be positioned in the proper location.

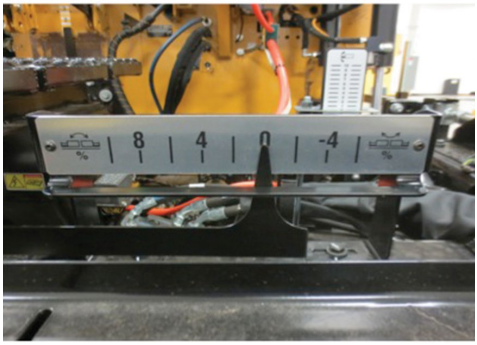
	Loose Mat	Tow Point Position
Deep Depth Paving 18-30 cm (7-12 in)	18 - 23 cm (7-9 in)	0-2.5 cm (0-1 in)
	23 - 30 cm (9-12 in)	2.5 - 7.5 cm (1-3 in)



STEP 3: Set Paving Width

With the screed still raised, move the extenders out to the proper paving width. When setting this up you will want to keep in mind the main screed width then add the additional width to meet the job specification. Each extender has a scale which shows how far the extender is moved out from the main screed.

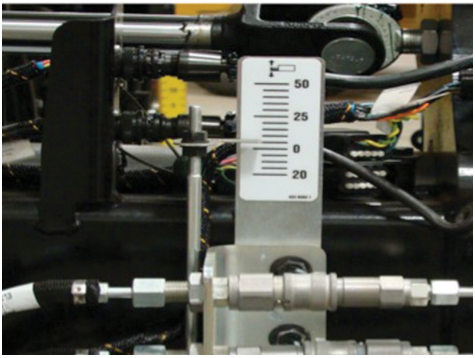
It is good paving practice to set equal extender width on each side whenever possible so the forces against each extender are equal. For example, if the job specifications are a total width of 12 ft (3.66 m), on the SE60 V / SE60 V XW the main screed measures 9 ft 10 in or 3 m, to achieve the proper width move the left and right extenders out an equal amount of 13 in (0.33 m) on each side. This will achieve the proper width required.



STEP 4: Set The Crown

Set the main screed crown to the specifications of the job.

Use the crown indicator scale located on the right side of the screed to set the crown. Before using the indicator scale, be sure that it is properly calibrated to the screed.



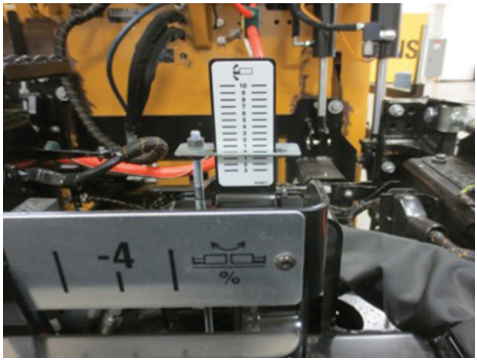
STEP 5: Set Extender Height

Set the height of the extenders so they match the height of the main screed. When the extender height is right, there are no transition marks in the mat.

Extender height is adjusted with a push button in each screed control box on vibratory screeds.

The SE series of screeds normally run with a 5 mm (3/16 in) nose-up angle of attack. Therefore, on a screed with rear mounted extenders, set the extender height 5 mm (3/16 in) above the zero point on the scale. Do this for both left and right extenders.

Again, make sure that the height indicators have been calibrated to the extenders.

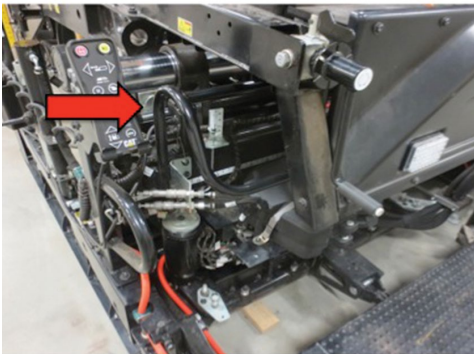


STEP 6: Set Extender Slope

Set the slope of the extenders to the specifications of the job.

If there is no requirement for extender slope, set the indicator at zero on both sides.

Again, make sure that the slope indicators have been calibrated to the extenders.



STEP 7: Lower The Screed

In this step we first want to raise the end gates to the proper position to ensure that the screed rests on the starting reference and not the end gates.

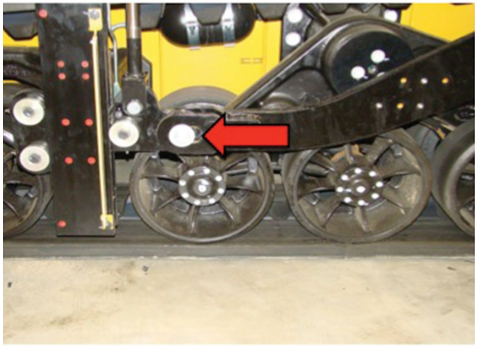
Then select a starting reference whose thickness matches the uncompacted mat depth that will be placed by the screed. Normally, the mat will compact about 6 mm (1/4 in per inch) per 25 mm (1 in) of mat depth. For example, to get a 50 mm (2 in) compacted mat, use a starting reference which is 64 mm (2-1/2 in) thick. Use two references. Position them so they are under the pivot of the extender and completely supporting both the main screed and the screed extender from front to back. The normal length of the starting references are between 0.9 - 1.2 m (3 - 4 ft).

NOTE: With the extenders beyond the halfway point, an additional starting reference for the extender may be required.



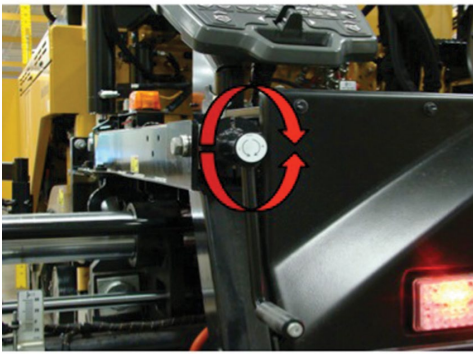
STEP 7: Lower The Screed ***(Continued)***

After placing your starting references, lower the screed by pressing the screed float button, making sure that the green light is illuminated next to the button. The screed is now in float and should lower to the references.



STEP 7: Lower The Screed *(Continued)*

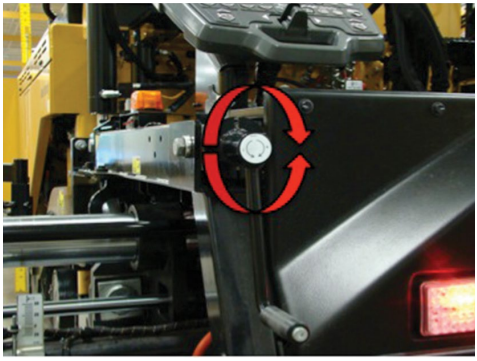
Step 7 also includes removing the slack from the tow arm. Move the machine forward until the tow arm roller contacts the tow point frame and both sides of the screed begin to move forward.



STEP 8: Nulling The Screed

Use the manual depth control cranks to null the screed. Nulling the screed should be done on one side of the screed at a time. Turn the crank in either direction until no resistance is felt. This ensures that the full weight of the screed is supported by the starting reference. Move to the other side and follow the same nulling procedure. Be sure the crank turns freely. Then, go back to the other side. Make sure the crank still turns freely.

NOTE: Repeat this until both sides have no resistance.

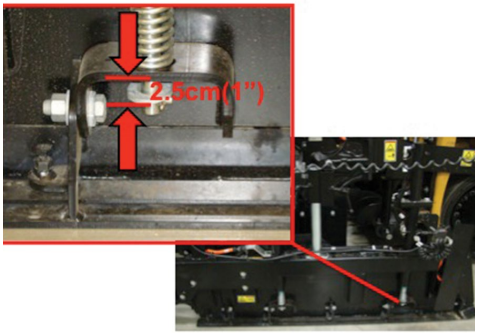


STEP 8: Nulling The Screed *(Continued)*

Then, turn the depth control crank in the direction that increases mat thickness until resistance is felt.

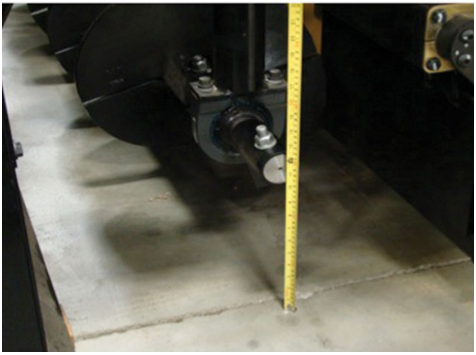
On the other side, turn the crank until you feel resistance.

On other types of screeds, you may have to turn the depth control cranks one or more revolutions to set the angle of attack. Follow the manufacturer's guidelines for the screed you are setting up.



STEP 9: Position The Endgates

End gates on both ends of the screed retain material at the proper width. They float on the grade. First, lower the end gate until it touches the grade. On manual gates, raise or lower the spring tension to create 2.54 cm (1 in) space between washer and bracket to allow for the end gates to move up and down over any grade irregularities. On the hydraulic power end gates lower them to the grade and press the “auto” button or raise even with bottom of the extender plate depending on the requirements.



STEP 10: Set The Auger Height

The height of the augers in relation to the depth of the uncompacted mat has an effect on the texture and finish of the mat. If the augers are too low, you will probably see an open texture and maybe material segregation. If the augers are too high, the head of material will likely be too high which will cause the screed to climb. Different mixes react differently to auger height adjustment, but as a rule, set the auger height at least 5 cm (2 in) above the height of the uncompacted mat. From the bottom of the auger to the center of the auger shaft is 20 cm (8 in). Now add 5 cm (2 in) to that. Then add the thickness of mat to be paved. The total is the distance from the centerline of the auger shaft to the grade. Adjust the auger up or down until the dimension is reached. Fine tune the auger height after paving starts if the mix is very coarse or very tender.



STEP 11: Set Feeder Sensor Position

Position the paddle type feeder sensors so they are sensing the active pile of material about 46 cm (18 in) away from the last auger segment. With the paddle-type sensor, the paddle arm should be at a 45 degree angle at the 46 cm (18 in) distance.